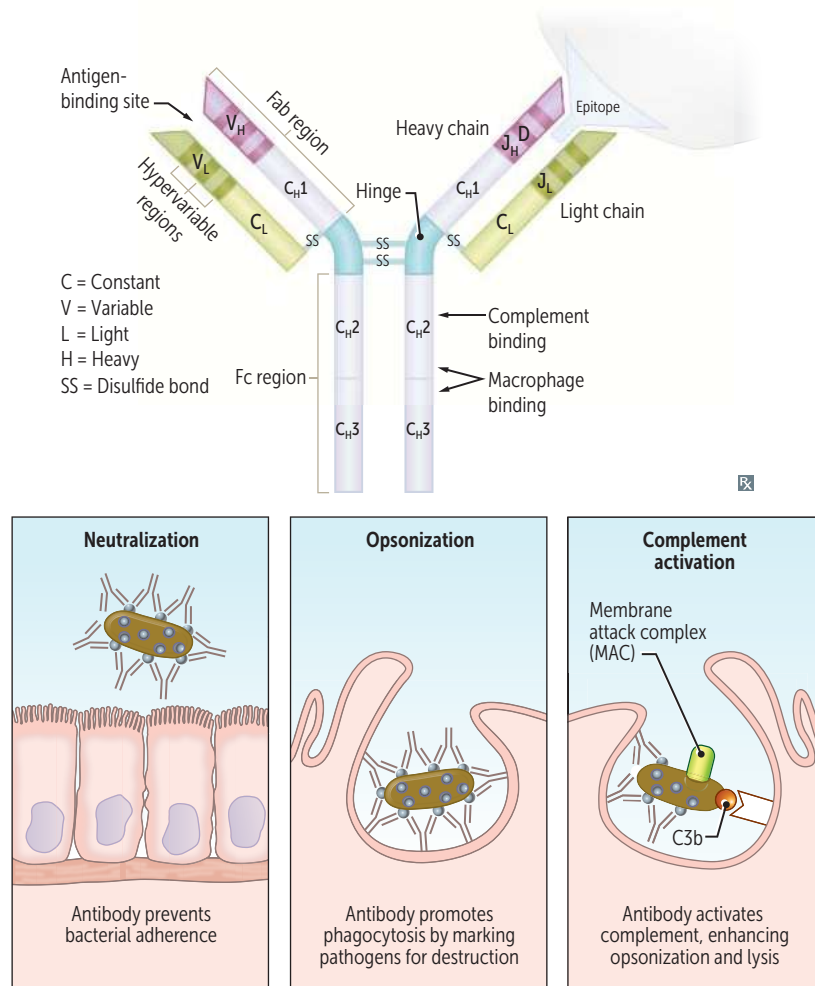


► IMMUNOLOGY—IMMUNE RESPONSES

Antibody structure and function

Fab fragment consisting of light (L) and heavy (H) chains recognizes antigens. Fc region of IgM and IgG fixes complement. Heavy chain contributes to Fc and Fab regions. Light chain contributes only to Fab region.

**Fab:**

- **F**ragment, **a**ntigen **b**inding
- Determines idotype: unique antigen-binding pocket; only 1 antigenic specificity expressed per B cell

Fc (5 C's):

- **C**onstant
- **C**arboxy terminal
- **C**omplement binding
- **C**arbohydrate side chains
- **C**onfers (determines) isotype (IgM, IgD, etc)

Generation of antibody diversity (antigen independent)

1. Random recombination of VJ (light-chain) or V(D)J (heavy-chain) genes by RAG1 and RAG2
2. Random addition of nucleotides to DNA during recombination by terminal deoxynucleotidyl transferase (TdT)
3. Random combination of heavy chains with light chains

Generation of antibody specificity (antigen dependent)

4. Somatic hypermutation and affinity maturation (variable region)
5. Isotype switching (constant region)

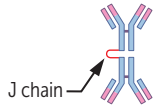
Immunoglobulin isotypes

All isotypes can exist as monomers. Mature, naïve B cells prior to activation express IgM and IgD on their surfaces. They may differentiate in germinal centers of lymph nodes by isotype switching (gene rearrangement; induced by cytokines and CD40L) into plasma cells that secrete IgA, IgG, or IgE. “For B cells, IgM_{om} and IgD_{ad} mature to plasma cells as they AGE.”

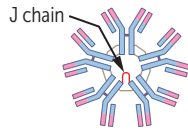
Affinity refers to the individual antibody-antigen interaction, while avidity describes the cumulative binding strength of all antibody-antigen interactions in a multivalent molecule.

IgG

Main antibody in 2° response to an antigen. Most abundant isotype in serum. Fixes complement, opsonizes bacteria, neutralizes bacterial toxins and viruses. Only isotype that crosses the placenta (provides infants with passive immunity that starts to wane after birth). “IgG Greets the Growing fetus.” Associated with warm autoimmune hemolytic anemia (“warm weather is Good!”).

IgA

Prevents attachment of bacteria and viruses to mucous membranes; does not fix complement. Monomer (in circulation) or dimer (with J chain when secreted). Crosses epithelial cells by transcytosis. Produced in GI tract (eg, by Peyer patches) and protects against gut infections (eg, *Giardia*). Most produced antibody overall, but has lower serum concentrations. Released into secretions (tears, saliva, mucus) and breast milk. Picks up secretory component from epithelial cells, which protects the Fc portion from luminal proteases.

IgM

First antibody to be produced during an immune response. Fixes complement. Antigen receptor on the surface of B cells. Monomer on B cell, pentamer with J chain when secreted. Pentamer enables avid binding to antigen while humoral response evolves. Associated with cold autoimmune hemolytic anemia.

IgD

Expressed on the surface of mature, naïve B cells. Normally, low levels are detectable in serum.

IgE

Binds mast cells and basophils; cross-links when exposed to allergen, mediating immediate (type I) hypersensitivity through release of inflammatory mediators such as histamine. Contributes to immunity to parasites by activating Eosinophils.

Antigen type and memory**Thymus-independent antigens**

Antigens lacking a peptide component (eg, lipopolysaccharides from gram $\ominus$ bacteria); cannot be presented by MHC to T cells. Weakly immunogenic; vaccines often require boosters and adjuvants (eg, capsular polysaccharide subunit of *Streptococcus pneumoniae* PPSV23 vaccine).

Thymus-dependent antigens

Antigens containing a protein component (eg, diphtheria toxoid). Class switching and immunologic memory occur as a result of direct contact of B cells with Th cells.

Complement

System of hepatically synthesized plasma proteins that play a role in innate immunity and inflammation. Membrane attack complex (MAC) defends against gram $\ominus$ bacteria. The CH_{50} test is used to screen for activation of the classical complement pathway.

ACTIVATION PATHWAYS

Classic—IgG or IgM mediated.

Alternative—bacterial products.

Lectin—mannose or other sugars on microbe surface.

General Motors makes classic cars.

FUNCTIONS

C3b—opsonization.

C3a, C4a, C5a—anaphylaxis.

C5a—neutrophil chemotaxis.

C5b-9 (MAC)—cytolysis.

C3b binds to lipopolysaccharides on bacteria.

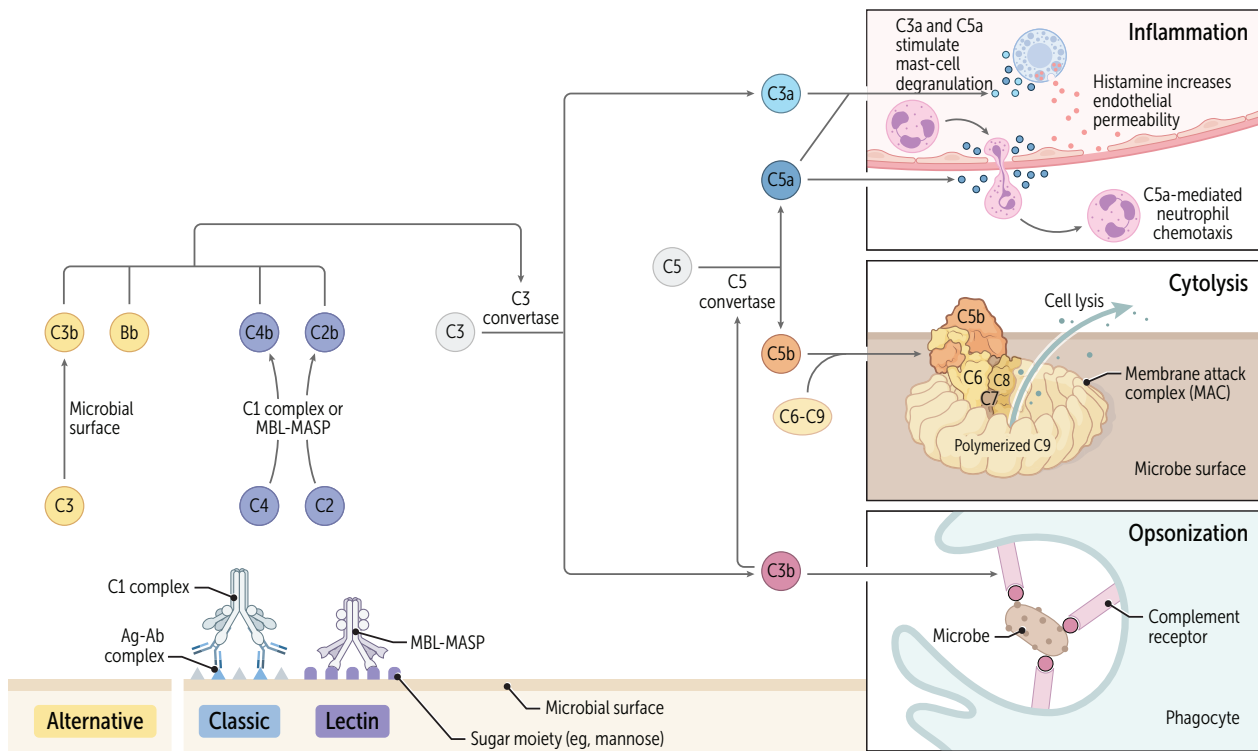
MAC complex is important for neutralizing *Neisseria* species. Deficiency results in recurrent infection.

Get “Neis” (nice) Big MACs from 5-9 pm.

Opsonin (Greek) = to prepare for eating.

Opsonins—C3b and IgG are the two 1° opsonins in bacterial defense; enhance phagocytosis. C3b also helps clear immune complexes.

Inhibitors—decay-accelerating factor (DAF, also called CD55) and C1 inhibitor (formerly called C1 esterase inhibitor) help prevent complement activation on self cells (eg, RBCs).



Complement disorders**Complement protein deficiencies**

Early complement deficiencies (C1–C4) ↑ risk of severe, recurrent pyogenic sinus and respiratory tract infections. C3b used in clearance of antigen-antibody complexes → ↑ risk of **SLE** (think **SLEarly**).

Terminal complement deficiencies (C5–C9) ↑ susceptibility to recurrent *Neisseria* bacteremia.

Complement regulatory protein deficiencies

C1 inhibitor deficiency Causes hereditary angioedema due to unregulated activation of kallikrein → ↑ bradykinin. Characterized by ↓ C4 levels. ACE inhibitors are contraindicated (also ↑ bradykinin).

Paroxysmal nocturnal hemoglobinuria A defect in the *PIGA* gene prevents the formation of glycosylphosphatidylinositol (GPI) anchors for complement inhibitors, such as decay-accelerating factor (DAF/CD55) and membrane inhibitor of reactive lysis (MIRL/CD59). Causes complement-mediated intravascular hemolysis → ↓ haptoglobin, dark urine **A**.



Can cause atypical venous thrombosis (eg, Budd-Chiari syndrome; portal vein, cerebral, or dermal thrombosis).

Treatment: eculizumab (anti-C5 antibody; inhibits terminal complement system and MAC formation).

Important cytokines Acute (IL-1, IL-6, TNF- α), then recruit (IL-8, IL-12).

Secreted by macrophages

Interleukin-1	Causes fever, acute inflammation. Activates endothelium to express adhesion molecules. Induces chemokine secretion to recruit WBCs. Also called osteoclast-activating factor.	“ Hot T-bone stEAK ”: IL-1: fever (hot). IL-2: stimulates T cells. IL-3: stimulates bone marrow. IL-4: stimulates Ig E production. IL-5: stimulates Ig A production. IL-6: stimulates a K ute-phase protein production.
Interleukin-6	Causes fever and stimulates production of acute-phase proteins.	
Tumor necrosis factor-α	Activates endothelium. Causes WBC recruitment, vascular leak.	Causes cachexia in malignancy. Maintains granulomas in TB. IL-1, IL-6, TNF- α can mediate fever and sepsis.
Interleukin-8	Major chemotactic factor for neutrophils.	“ Clean up on aisle 8. ” Neutrophils are recruited by IL-8 to clear infections.
Interleukin-12	Induces differentiation of T cells into Th1 cells. Activates NK cells.	Facilitates granuloma formation in TB.

Secreted by T cells

Interleukin-2	Stimulates growth of helper, cytotoxic, and regulatory T cells, and NK cells.	
Interleukin-3	Supports growth and differentiation of bone marrow stem cells. Functions like GM-CSF.	Stimulates proliferation of eosinophils, basophils, neutrophils, monocytes.

From Th1 cells

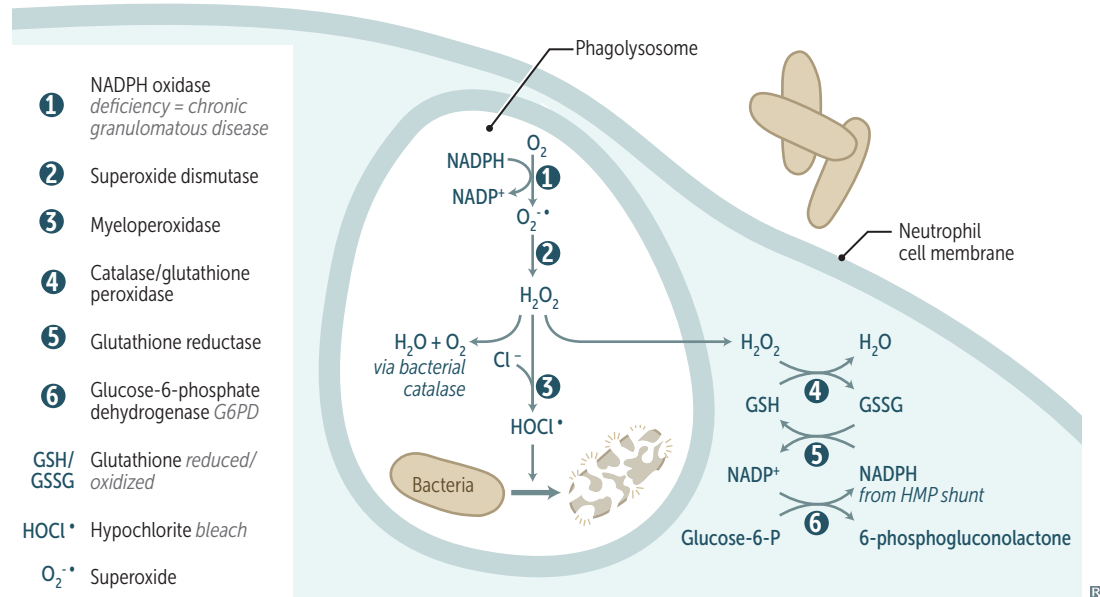
Interferon-γ	Secreted by NK cells and T cells in response to antigen or IL-12 from macrophages; stimulates macrophages to kill phagocytosed pathogens. Inhibits differentiation of Th2 cells. Induces Ig G isotype switching in B cells.	Increases MHC expression and antigen presentation by all cells. Activates macrophages to induce granuloma formation.
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From Th2 cells

Interleukin-4	Induces differentiation of T cells into Th (helper) 2 cells. Promotes growth of B cells. Enhances class switching to Ig E and Ig G .	Ain't too proud 2 BEG 4 help .
Interleukin-5	Promotes growth and differentiation of B cells. Enhances class switching to Ig A . Stimulates growth and differentiation of E osinophils.	I have 5 BAEs .
Interleukin-10	Attenuates inflammatory response. Decreases expression of MHC class II and Th1 cytokines. Inhibits activated macrophages and dendritic cells. Also secreted by regulatory T cells.	TGF- β and IL- 10 both attenuate the immune response.
Interleukin-13	Promotes IgE production by B cells. Induces alternative macrophage activation.	Interleukin thirt EE n promotes Ig E .

Respiratory burst

Also called oxidative burst. Involves the activation of the phagocyte NADPH oxidase complex (eg, in neutrophils, monocytes), which utilizes O_2 as a substrate. Plays an important role in the immune response → rapid release of reactive oxygen species (ROS). NADPH plays a role in both the creation and neutralization of ROS. Myeloperoxidase contains a blue-green, heme-containing pigment that gives sputum its color. **NO Safe Microbe** (NADPH Oxidase → Superoxide dismutase → Myeloperoxidase).



Phagocytes of patients with CGD can utilize H_2O_2 generated by invading organisms and convert it to ROS. Patients are at ↑ risk for infection by catalase ⊕ species (eg, *S aureus*, *Aspergillus*) capable of neutralizing their own H_2O_2 , leaving phagocytes without ROS for fighting infections.

Pyocyanin of *P aeruginosa* generates ROS to kill competing pathogens. Oxidative burst leads to release of lysosomal enzymes.

Type I interferons

IFN- α , IFN- β .

MECHANISM

A part of innate host defense, **interferons interfere** with both RNA and DNA viruses. Cells infected with a virus synthesize these glycoproteins, which act on local cells, priming them for viral defense by downregulating protein synthesis to resist potential viral replication and by upregulating MHC expression to facilitate recognition of infected cells. Also play a major role in activating antitumor immunity.

CLINICAL USE

Chronic HBV, Kaposi sarcoma, hairy cell leukemia, condyloma acuminatum, renal cell carcinoma, malignant melanoma, multiple sclerosis.

ADVERSE EFFECTS

Flulike symptoms, depression, neutropenia, myopathy, interferon-induced autoimmunity.

Cell surface proteins

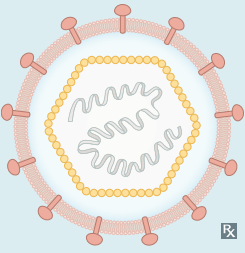
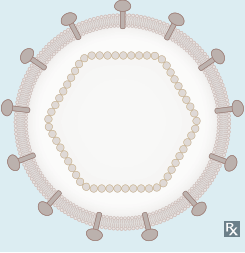
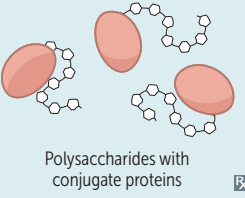
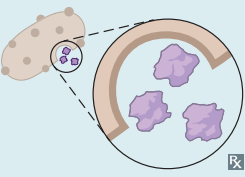
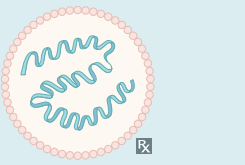
T cells	TCR (binds antigen-MHC complex), CD3 (associated with TCR for signal transduction), CD28 (binds B7 on APC)	
Helper T cells	CD4, CD40L, CXCR4/CCR5 (coreceptors for HIV)	
Cytotoxic T cells	CD8	
Regulatory T cells	CD4, CD25	
B cells	Ig (binds antigen), CD19, CD20, CD21 (receptor for Epstein-Barr virus), CD40, MHC II, B7 (CD80/86)	Must be 21 to drink at a Barr
NK cells	CD16 (binds Fc of IgG), CD56 (suggestive marker for NK cells)	
Macrophages	CD14 (receptor for PAMPs [eg, LPS]), CD40, CCR5, MHC II, B7, Fc and C3b receptors (enhanced phagocytosis)	
Hematopoietic stem cells	CD34	

Passive vs active immunity

	Passive	Active
MEANS OF ACQUISITION	Receiving preformed antibodies	Exposure to exogenous antigens
ONSET	Rapid	Slow
DURATION	Short span of antibodies (half-life = 3 weeks)	Long-lasting protection (memory)
EXAMPLES	IgA in breast milk, maternal IgG crossing placenta, antitoxin, humanized monoclonal antibody	Natural infection, vaccines, toxoid
NOTES	IVIG and other immune globulin preparations can be administered to provide temporary but specific passive immunity to a target pathogen	Combined passive and active immunizations can be given for hepatitis B or rabies exposure

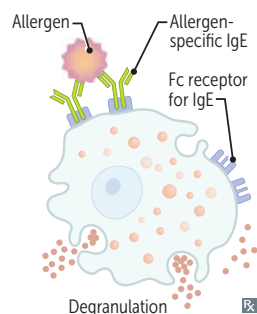
Vaccination

Induces an active immune response (humoral and/or cellular) to specific pathogens.

VACCINE TYPE	DESCRIPTION	PROS/CONS	EXAMPLES
Live attenuated vaccine 	Microorganism rendered nonpathogenic but retains capacity for transient growth within inoculated host. Certain live vaccines (MMR, varicella) may be given to people living with HIV who have a CD4+ cell count ≥ 200 cells/mm ³ in consultation with a specialist in infectious disease or immunology.	Pros: induces cellular and humoral responses. Induces strong, often lifelong immunity. Cons: may revert to virulent form. Contraindicated in pregnancy and patients with immunodeficiency.	Adenovirus (nonattenuated, given to military recruits), typhoid (Ty21a, oral), polio (Sabin), varicella (chickenpox), smallpox, BCG, yellow fever, influenza (intranasal), MMR, rotavirus. “Attention teachers! Please vaccinate small, Beautiful young infants with MMR routinely!”
Killed or inactivated vaccine 	Pathogen is inactivated by heat or chemicals. Maintaining epitope structure on surface antigens is important for immune response. Mainly induces a humoral response.	Pros: safer than live vaccines. Cons: weaker cell-mediated immune response; mainly induces a humoral response. Booster shots usually needed.	Hepatitis A, Typhoid (Vi polysaccharide, intramuscular), Rabies, Influenza (intramuscular), Polio (SalK). A TRIP could Kill you.
Subunit, recombinant, polysaccharide, and conjugate 	All use specific antigens that best stimulate the immune system.	Pros: targets specific epitopes of antigen; lower chance of adverse reactions. Cons: expensive; weaker immune response.	HBV (antigen = HBsAg), HPV, acellular pertussis (aP), <i>Neisseria meningitidis</i> (various strains), <i>Streptococcus pneumoniae</i> (PPSV23 polysaccharide primarily T-cell-independent response; PCV13, PCV15, and PCV20 polysaccharide produces T-cell-dependent response), Hib, herpes zoster.
Toxoid 	Denatured bacterial toxin with an intact receptor binding site. Stimulates immune system to make antibodies without potential for causing disease.	Pros: protects against the bacterial toxins. Cons: antitoxin levels decrease with time, thus booster shots may be needed.	<i>Clostridium tetani</i> , <i>Corynebacterium diphtheriae</i> .
mRNA 	A lipid nanoparticle delivers mRNA, causing cells to synthesize foreign protein (eg, spike protein of SARS-CoV-2).	Pros: high efficacy; induces cellular and humoral immunity. Safe in pregnancy. Cons: local and transient systemic (fatigue, headache, myalgia) reactions are common. Rare myocarditis, pericarditis particularly in young males.	SARS-CoV-2.

Hypersensitivity types Four types: Anaphylactic and atopic (type I), antibody-mediated (type II), immune complex (type III), cell-mediated (type IV). Types I, II, and III are all antibody-mediated.

Type I hypersensitivity



Anaphylactic and atopic—two phases:

- Immediate (minutes): antigen crosslinks preformed IgE on presensitized mast cells → immediate degranulation → release of histamine (a vasoactive amine), tryptase (marker of mast cell activation), and leukotrienes.
- Late (hours): chemokines (attract inflammatory cells, eg, eosinophils) and other mediators from mast cells → inflammation and tissue damage.

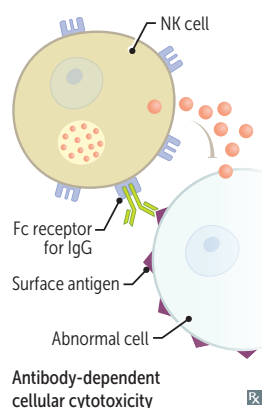
First (type) and **F**ast (anaphylaxis).

Test: skin test or blood test (ELISA) for allergen-specific IgE.

Example:

- Anaphylaxis (eg, food, drug, or bee sting allergies)
- Allergic asthma

Type II hypersensitivity



Antibodies bind to cell-surface antigens or extracellular matrix → cellular destruction, inflammation, and cellular dysfunction.

Cellular destruction—cell is opsonized (coated) by antibodies, leading to either:

- Phagocytosis and/or activation of complement system.
- NK cell killing (antibody-dependent cellular cytotoxicity).

Inflammation—binding of antibodies to cell surfaces → activation of complement system and Fc receptor-mediated inflammation.

Cellular dysfunction—antibodies bind to cell-surface receptors → abnormal blockade or activation of downstream process.

Direct Coombs test—detects antibodies attached **directly** to the RBC surface.

Indirect Coombs test—detects presence of unbound antibodies in the serum.

Examples:

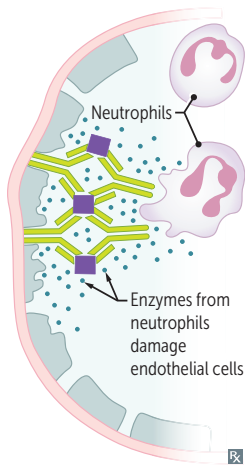
- Autoimmune hemolytic anemia (including drug-induced form)
- Immune thrombocytopenia
- Transfusion reactions
- Hemolytic disease of the newborn

Examples:

- Goodpasture syndrome
- Rheumatic fever
- Hyperacute transplant rejection

Examples:

- Myasthenia gravis
- Graves disease
- Pemphigus vulgaris

Hypersensitivity types (continued)**Type III hypersensitivity**

Immune complex—antigen-antibody (mostly IgG) complexes activate complement, which attracts neutrophils; neutrophils release lysosomal enzymes.

Can be associated with vasculitis and systemic manifestations.

Serum sickness—the prototypic immune complex disease. Antibodies to foreign proteins are produced and 1–2 weeks later, antibody-antigen complexes form and deposit in tissues → complement activation → inflammation and tissue damage (↓ serum C3, C4).

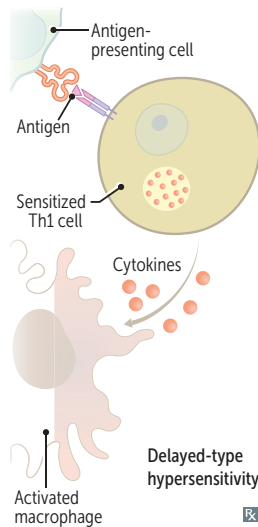
Arthus reaction—a local subacute immune complex-mediated hypersensitivity reaction. Intradermal injection of antigen into a presensitized (has circulating IgG) individual leads to immune complex formation in the skin (eg, enhanced local reaction to a booster vaccination). Characterized by edema, fibrinoid necrosis, activation of complement.

In type **III** reaction, imagine an immune complex as **3** things stuck together: antigen-antibody-complement.

Examples:

- SLE
- Reactive arthritis
- Polyarteritis nodosa
- Poststreptococcal glomerulonephritis
- IgA vasculitis

Fever, urticaria, arthralgia, proteinuria, lymphadenopathy occur 1–2 weeks after antigen exposure. Serum sickness–like reactions are associated with some drugs (may act as haptens, eg, penicillin, monoclonal antibodies) and infections (eg, hepatitis B).

Type IV hypersensitivity

Two mechanisms, each involving T cells:

1. Direct cell cytotoxicity: CD8+ cytotoxic T cells kill targeted cells.
2. Inflammatory reaction: effector CD4+ T cells recognize antigen and release inflammation-inducing cytokines (shown in illustration).

Response does not involve antibodies (vs types I, II, and III).

Examples:

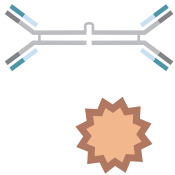
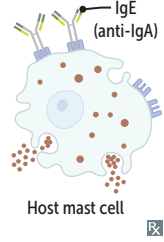
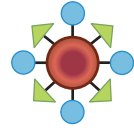
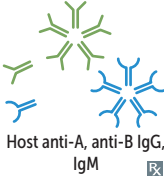
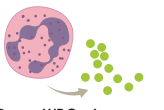
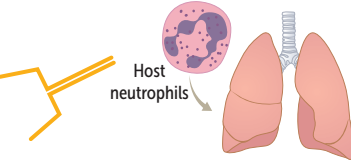
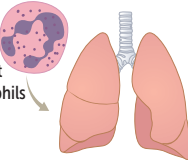
- Contact dermatitis (eg, poison ivy, nickel allergy)
- Drug reaction with eosinophilia and systemic symptoms (DRESS)
- Graft-versus-host disease

Tests: PPD for TB infection; patch test for contact dermatitis; *Candida* skin test for T cell immune function.

4T's: **T** cells, **T**ransplant rejections, **T**B skin tests, **T**ouching (contact dermatitis).

Fourth (type) and **last** (delayed).

Immunologic blood transfusion reactions

TYPE	PATHOGENESIS	TIMING	CLINICAL PRESENTATION	DONOR BLOOD	HOST BLOOD
Allergic/ anaphylactic reaction	Type I hypersensitivity reaction against plasma proteins in transfused blood IgA-deficient individuals should receive blood products without IgA	Within minutes to 2–3 hr (due to release of preformed inflammatory mediators in degranulating mast cells)	Allergies: urticaria, pruritus Anaphylaxis: wheezing, hypotension, respiratory arrest, shock	 Donor plasma proteins, including IgA	 Host mast cell 
Acute hemolytic transfusion reaction	Type II hypersensitivity reaction Typically causes intravascular hemolysis (ABO blood group incompatibility)	During transfusion or within 24 hr (due to preformed antibodies)	Fever, hypotension, tachypnea, tachycardia, flank pain, hemoglobinuria (intravascular), jaundice (extravascular)	 Donor RBC with A and/or B group antigens	 Host anti-A, anti-B IgG, IgM 
Febrile nonhemolytic transfusion reaction	Cytokines created by donor WBCs accumulate during storage of blood products Reactions prevented by leukoreduction of blood products	Within 1–6 hr (due to preformed cytokines)	Fever, headaches, chills, flushing More common in children	 Donor WBC releases preformed cytokines 	
Transfusion-related acute lung injury	Two-hit mechanism: ▪ Neutrophils are sequestered and primed in pulmonary vasculature due to recipient risk factors ▪ Neutrophils are activated by a product (eg, antileukocyte antibodies) in the transfused blood and release inflammatory mediators → ↑ capillary permeability → pulmonary edema	Within minutes to 6 hr	Respiratory distress, noncardiogenic pulmonary edema	 Donor antileukocyte antibody	 Host neutrophils 
Delayed hemolytic transfusion reaction	Anamnestic response to a foreign antigen on donor RBCs (Rh [D] or other minor blood group antigens) previously encountered by recipient Typically causes extravascular hemolysis	Onset over 24 hr Usually presents within 1–2 wk (due to slow destruction by reticuloendothelial system)	Generally self limited and clinically silent Mild fever, hyperbilirubinemia		

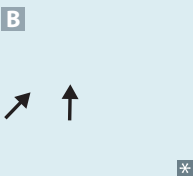
Autoantibodies

AUTOANTIBODY	ASSOCIATED DISORDER
Anti-postsynaptic ACh receptor	Myasthenia gravis
Anti-presynaptic voltage-gated Ca^{2+} channel	Lambert-Eaton myasthenic syndrome
Anti- β_2 glycoprotein I	Antiphospholipid syndrome
Antinuclear (ANA)	Nonspecific screening antibody, often associated with SLE
Anticardiolipin, lupus anticoagulant	SLE, antiphospholipid syndrome
Anti-dsDNA, anti-Smith	SLE
Antihistone	Drug-induced lupus
Anti-U1 RNP (ribonucleoprotein)	Mixed connective tissue disease
Rheumatoid factor (IgM antibody against IgG Fc region), anti-cyclic citrullinated peptide (anti-CCP, more specific)	Rheumatoid arthritis
Anti-Ro/ SSA , anti-La/ SSB	Sjögren syndrome
Anti- Scl -70 (anti-DNA topoisomerase I)	Scleroderma (diffuse)
Anticentromere	Limited scleroderma (CREST syndrome)
Antisynthetase (eg, anti-Jo-1), anti-SRP, anti-helicase (anti-Mi-2)	Polymyositis, dermatomyositis
Antimitochondrial	1° biliary cholangitis
Anti-smooth muscle, anti-liver/kidney microsomal-1	Autoimmune hepatitis
Myeloperoxidase-antineutrophil cytoplasmic antibody (MPO-ANCA)/perinuclear ANCA (p-ANCA)	Microscopic polyangiitis, eosinophilic granulomatosis with polyangiitis, ulcerative colitis, 1° sclerosing cholangitis
PR3-ANCA/cytoplasmic ANCA (c-ANCA)	Granulomatosis with polyangiitis
Anti-phospholipase A ₂ receptor	1° membranous nephropathy
Anti-hemidesmosome	Bullous pemphigoid
Anti-desmoglein (anti-desmosome)	Pemphigus vulgaris
Antithyroglobulin, antithyroid peroxidase (antimicrosomal)	Hashimoto thyroiditis
Anti-TSH receptor	Graves disease
IgA anti-endomysial, IgA anti-tissue transglutaminase, IgA and IgG deamidated gliadin peptide	Celiac disease
Anti-glutamic acid decarboxylase, islet cell cytoplasmic antibodies	Type 1 diabetes mellitus
Antiparietal cell, anti-intrinsic factor	Pernicious anemia
Anti-glomerular basement membrane	Goodpasture syndrome

Immunodeficiencies

DISEASE	DEFECT	PRESENTATION	FINDINGS
B-cell disorders			
X-linked (Bruton) agammaglobulinemia	Defect in BTK , a tyrosine kinase gene → no B -cell maturation; X-linked recessive (↑ in Boys)	Recurrent bacterial and enteroviral infections after 6 months (↓ maternal IgG)	Absent B cells in peripheral blood, ↓ Ig of all classes. Absent/scanty lymph nodes and tonsils (1° follicles and germinal centers absent) → live vaccines contraindicated
Selective IgA deficiency	May be familial or sporadic Most common 1° immunodeficiency May also arise 2° to certain viral infections or medications	Majority A symptomatic Can see A irway and GI infections, A utoimmune disease, A topy, A naphylaxis to IgA in blood products	↓ IgA with normal IgG, IgM levels ↑ susceptibility to giardiasis Can cause false-negative celiac disease test and false-positive serum pregnancy test
Common variable immunodeficiency	Defect in B-cell differentiation. Cause unknown in most cases	May present in childhood but usually diagnosed after puberty ↑ risk of autoimmune disease, bronchiectasis, lymphoma, sinopulmonary infections	↓ plasma cells, ↓ immunoglobulins
T-cell disorders			
Thymic aplasia	22q11 microdeletion; failure to develop 3rd and 4th pharyngeal pouches → absent thymus and parathyroids DiGeorge syndrome —thymic, parathyroid, cardiac defects Velocardiofacial syndrome —palate, facial, cardiac defects	CATCH-22 : Cardiac defects (conotruncal abnormalities [eg, tetralogy of Fallot, truncus arteriosus]), A bnormal facies, T hymic hypoplasia → T-cell deficiency (recurrent viral/fungal infections), C left palate, H ypocalcemia 2° to parathyroid aplasia → tetany	↓ T cells, ↓ PTH, ↓ Ca ²⁺ Thymic shadow absent on CXR
IL-12 receptor deficiency	↓ Th1 response; autosomal recessive	Disseminated mycobacterial and fungal infections; may present after administration of BCG vaccine	↓ IFN-γ Most common cause of Mendelian susceptibility to mycobacterial diseases (MSMD)
Autosomal dominant hyper-IgE syndrome (Job syndrome)	Deficiency of Th17 cells due to STAT3 mutation → impaired recruitment of neutrophils to sites of infection	Cold (noninflamed) staphylococcal A bscesses, retained B aby teeth, C oarse facies, D ermatologic problems (eczema), ↑ Ig E , bone F ractures from minor trauma	↑ IgE ↑ eosinophils Learn the ABCDEF 's to get a Job STAT!
Chronic mucocutaneous candidiasis	T-cell dysfunction Impaired cell-mediated immunity against <i>Candida</i> sp Classic form caused by defects in AIRE	Persistent noninvasive <i>Candida albicans</i> infections of skin and mucous membranes	Absent in vitro T-cell proliferation in response to <i>Candida</i> antigens Absent cutaneous reaction to <i>Candida</i> antigens

Immunodeficiencies (continued)

DISEASE	DEFECT	PRESENTATION	FINDINGS
B- and T-cell disorders			
Severe combined immunodeficiency	Several types including defective IL-2R gamma chain (most common, X-linked recessive); adenosine deaminase deficiency (autosomal recessive); RAG mutation → VDJ recombination defect	Failure to thrive, chronic diarrhea, thrush Recurrent viral, bacterial, fungal, and protozoal infections	↓ T-cell receptor excision circles (TRECs) Part of newborn screening for SCID Absence of thymic shadow (CXR), germinal centers (lymph node biopsy), and T cells (flow cytometry)
Ataxia-telangiectasia 	Defects in ATM gene → failure to detect DNA damage → failure to halt progression of cell cycle → mutations accumulate; autosomal recessive	Triad: cerebellar defects (Ataxia), spider Angiomas (telangiectasia A), IgA deficiency ↑↑ sensitivity to radiation (limit x-ray exposure)	↑ AFP ↓ IgA, IgG, and IgE Lymphopenia, cerebellar atrophy ↑ risk of lymphoma and leukemia
Hyper-IgM syndrome	Most commonly due to defective CD40L on Th cells → class switching defect; X-linked recessive	Severe pyogenic infections early in life; opportunistic infection with <i>Pneumocystis</i> , <i>Cryptosporidium</i> , CMV	Normal or ↑ IgM ↓↓ IgG, IgA, IgE Failure to make germinal centers
Wiskott-Aldrich syndrome	WAS mutation; leukocytes and platelets unable to reorganize actin cytoskeleton → defective antigen presentation; X-linked recessive	WATER : Wiskott-Aldrich: Thrombocytopenia , Eczema , Recurrent (pyogenic) infections ↑ risk of autoimmune disease and malignancy	↓ to normal IgG, IgM ↑ IgE, IgA Fewer and smaller platelets
Phagocyte dysfunction			
Leukocyte adhesion deficiency (type 1)	Autosomal recessive defect in LFA-1 integrin (CD18) protein on phagocytes leads to impaired migration and chemotaxis by C5a, IL-8, and leukotriene B4	L ate separation (>30 days) of umbilical cord, a bsent pus, d ysfunctional neutrophils → recurrent skin and mucosal bacterial infections	↑ neutrophils in blood Absence of neutrophils at infection sites → impaired wound healing
Chédiak-Higashi syndrome 	Defect in lysosomal trafficking regulator gene (<i>LYST</i>) Microtubule dysfunction in phagosome-lysosome fusion; autosomal recessive	PLAIN : P rogressive neurodegeneration, L ymphohistiocytosis, A lbinism (partial), recurrent pyogenic I nfections, peripheral N europathy	Giant granules (B , arrows) in granulocytes and platelets Pancytopenia Mild coagulation defects
Chronic granulomatous disease	Defect of NADPH oxidase → ↓ reactive oxygen species (eg, superoxide) and ↓ respiratory burst in neutrophils; X-linked form most common	↑ susceptibility to catalase ⊕ organisms (granny's cats keep her positive) Recurrent infections and granulomas	Abnormal dihydrorhodamine (flow cytometry) test (↓ green fluorescence) Nitroblue tetrazolium dye reduction test (obsolete) fails to turn blue

Infections in immunodeficiency

PATHOGEN	↓ T CELLS	↓ B CELLS	↓ GRANULOCYTES	↓ COMPLEMENT
Bacteria	Sepsis	Encapsulated (Please SHINE my SKiS): <i>Pseudomonas aeruginosa</i> , <i>Streptococcus pneumoniae</i> , <i>Haemophilus Influenzae</i> type b, <i>Neisseria meningitidis</i> , <i>Escherichia coli</i> , <i>Salmonella</i> , <i>Klebsiella pneumoniae</i> , group B <i>Streptococcus</i>	Some Bacteria Produce No Serious granules: <i>Staphylococcus</i> , <i>Burkholderia cepacia</i> , <i>Pseudomonas aeruginosa</i> , <i>Nocardia</i> , <i>Serratia</i>	Encapsulated species with early complement deficiencies <i>Neisseria</i> with late complement (C5– C9) deficiencies
Viruses	CMV, EBV, JC virus, VZV, chronic infection with respiratory/GI viruses	Enteroviral encephalitis, poliovirus (live vaccine contraindicated)	N/A	N/A
Fungi/parasites	<i>Candida</i> (local), PCP, <i>Cryptococcus</i>	GI giardiasis (no IgA)	<i>Candida</i> (systemic), <i>Aspergillus</i> , <i>Mucor</i>	N/A

Note: B-cell deficiencies tend to produce recurrent bacterial infections, whereas T-cell deficiencies produce more fungal and viral infections.

Transplant rejection

TYPE OF REJECTION	ONSET	PATHOGENESIS	FEATURES
Hyperacute	Within minutes	Pre-existing recipient antibodies react to donor antigen (type II hypersensitivity reaction), activate complement	Widespread thrombosis of graft vessels (arrows within glomerulus A) → ischemia and fibrinoid necrosis Graft must be removed
Acute	Weeks to months	Cellular: CD8+ T cells and/or CD4+ T cells activated against donor MHCs (type IV hypersensitivity reaction) Humoral: similar to hyperacute, except antibodies develop after transplant (associated with C4d deposition)	Vasculitis of graft vessels with dense interstitial lymphocytic infiltrate B Prevent/reverse with immunosuppressants
Chronic	Months to years	CD4+ T cells respond to recipient APCs presenting donor peptides, including allogeneic MHC Both cellular and humoral components (type II and IV hypersensitivity reactions)	Dominated by arteriosclerosis C Recipient T cells react and secrete cytokines → proliferation of vascular smooth muscle, parenchymal atrophy, interstitial fibrosis Organ-specific examples: ▪ Chronic allograft nephropathy ▪ Bronchiolitis obliterans ▪ Accelerated atherosclerosis (heart) ▪ Vanishing bile duct syndrome
Graft-versus-host disease	Varies	Grafted immunocompetent T cells proliferate in the immunocompromised host and reject host cells with “foreign” proteins → severe organ dysfunction HLA mismatches ↑ the risk for GVHD Type IV hypersensitivity reaction	Maculopapular rash, jaundice, diarrhea, hepatosplenomegaly Usually in bone marrow and liver transplants (rich in lymphocytes) Potentially beneficial in bone marrow transplant for leukemia (graft-versus-tumor effect) For patients who are immunocompromised, irradiate blood products prior to transfusion to prevent GVHD

